

LXCE on ThinkSystem V4 servers

New features and enhancements

Lenovo

Overview

LXCE has the following new features and enhancements for the ThinkSystem V4 platform:

- Version updates to support ThinkSystem V4
 - OneCLI: 5.0.0
 - UpdateXpress: 5.0.0
 - BoMC: 14.0.0
- New OS support
 - Windows: Windows Server 2025
 - Linux: RHCE 8.10/9.4, SLES 15.6, Ubuntu 24.04

Always check the Lenovo XClarity Essentials OneCLI (OneCLI) website for the latest OneCLI information and User Guide.

- <https://datacentersupport.lenovo.com/us/en/documents/Invo-tcli>

Binary files

The following binary file names will be updated in accordance with the version change:

OneCLI Binary

- Invgy_utl_lxce_oneclixxx-5.0.0_windows_indiv.zip
- Invgy_utl_lxceb_oneclixxx-5.0.0_windows_indiv.exe
- Invgy_utl_lxce_oneclixxx-5.0.0_linux_indiv.tgz
- Invgy_utl_lxceb_oneclixxx-5.0.0_linux_indiv.bin
- Invgy_utl_lxcer_oneclixxx-5.0.0_linux_indiv.rpm

UpdateXpress Binary

- Invgy_utl_lxce_uxxxx-5.0.0_windows_indiv.zip
- Invgy_utl_lxce_uxxxx-5.0.0_linux_indiv.tgz

BoMC Binary

- Invgy_utl_lxce_bomcxxx-14.0.0_windows_indiv.exe
- Invgy_utl_lxce_bomcxxx-14.0.0_linux_indiv.bin

Note that file names now end with **indiv**.

LXCE major software features summary

LXCE has the following feature updates for the ThinkSystem V4 platform:

- OneCLI - LXUM (Lenovo XClarity Update Manager) replaces the legacy Bare Metal Update (BMU) on BHS
- OneCLI - In-band disk drive firmware update without a reboot
- OneCLI - Configuration with OneCLI
- OneCLI - Back up/restore the configuration with encryption
- OneCLI - Miscellaneous new OneCLI functions
- UpdateXpress - ThinkEdge server security feature
- UpdateXpress - Back up and restore system configuration settings
- BoMC - Support for the ST45 V3 FW update

OneCLI LXUM replacing the legacy BMU on BHS

LXUM is a core firmware component on the ThinkSystem V3 (EGS/Genoa) and V4 (BHS) platforms. It is used to boot a BMU environment via XCC to carry out in-band (IB) firmware updates. This includes OSs under a hypervisor or not on the list of supported OSs – for example, HDD or some GPU firmware updates. As the OneCLI command for firmware updates is still the same, from a service perspective, this feature has no impact.

Command:

```
OneCli update flash --dir pkg --  
platform -b  
USERID:PASSWORD@IPADDRESS --log 5
```

```
Target host may will be boot to Maintenance Mode OS for update in 00s, if you want stop it press ctrl+c.  
Start platform flashing...  
  
*****  
[00s]: 3 packages will be updating.  
Prepare the environment of updating...  
  
00B>Current flashing ID:lnvgy_fw_sysboard_sep.g1-88-11-0231_anyos_comp  
  
[1s]Uploading succeed [100%][=====]  
  
[16s]Task Completed [100%][=====]  
  
00B>Current flashing ID:lnvgy_fw_sysboard_sep.g1-8A-3-0231_anyos_comp  
  
[1s]Uploading succeed [100%][=====]  
  
[15s]Task Completed [100%][=====]  
  
00B>Current flashing ID:lnvgy_fw_lxum_eal505c-1.10_anyos_comp  
  
[12s]Uploading succeed [100%][=====]  
  
[32s]Task Completed [100%][=====]  
Will boot os and check packages whether has taken effect.  
Rebooting...  
BMU init: Starting boot to Maintenance Mode OS [ 7%][=====]
```


BMU image ~~not required~~ when using LXUM in V4

The old BMU and new LXUM update command is the same:

```
OneCli update flash --dir pkg --platform -b USERID:PASSWORD@IPADDRESS --log 5  
pkg: the local folder that contains the payload
```

With both V3 and V4 systems, the folder specified by `--dir pkg` must contain the payload and OneCLI platform package. The BMU image (`lnvgy_utl_lxce_bmu-xxx-xxx_anyos_32-64.iso`) is required for V3 systems. With V4 systems, the image could be included but is not required.

intc-lnvgy_dd_nic_net-j9cnv-00e2-intel-i40e-kmp-default-2....	6/19/2024 5:29 PM	JSON File
intc-lnvgy_dd_nic_net-j9cnv-00e2-intel-i40e-kmp-default-2....	6/19/2024 5:27 PM	JSON File
intc-lnvgy_dd_nic_net-j9cnv-00e2-intel-i40e-kmp-default-2....	6/19/2024 5:27 PM	JSON File
intc-lnvgy_dd_nic_net-j9cnv-00e2-intel-i40e-kmp-default-2....	6/19/2024 5:27 PM	JSON File
intc-lnvgy_dd_nic_net-j9cnv-00e2-intel-i40e-kmp-default-2....	6/19/2024 5:29 PM	JSON File
lnvgy_utl_lxce_bmu-1.0.4-4.0.18_anyos_32-64.iso		
lnvgy_utl_lxce_onecli01o-5.0.0_linux_x86-64.tgz	6/19/2024 5:27 PM	JSON File

Not required on V4

In-band disk drive firmware update without a reboot

Starting with version 5.0.0, OneCLI includes a drive firmware update utility to support IB disk drive firmware (FW) updates without a reboot.

- The drive FW package is required
- Both Windows and Linux are supported

Command:

```
./OneCli update flash -dir  
pkg --log 5
```

```
Start inband flashing...

Current flashing ID:lnvgy_fw_drives_all.stx.exos.15e900.146-146c-0122_anyos_comp
Option inband flash.....
[12s]Succeed [100%]=====

Current flashing ID:lnvgy_fw_drives_all.mu.7450-e2mu200-01e2_anyos_comp
Option inband flash.....
[1s]Succeed [100%]=====

Current flashing ID:lnvgy_fw_drives_all.stx.hdd.150-150d-00e2_anyos_comp
Option inband flash.....
[1m12s]Succeed [100%]=====

The firmware update utility lnvgy_fwutl_drives_all-j9c8p-0022_anyos_comp isn't needed, skip to install it.
The firmware update utility lnvgy_fwutl_drives_all-j9crb-0002_anyos_comp isn't needed, skip to install it.

Flash Results:
=====
| No. | Updatable Unit | Slot | Need Reboot | Result | Update ID |
=====
| 1 | ST900MP0146 | Drive_Bay | No reboot required | success | lnvgy_fw_drives_all.stx.exos.15e900.146-146c-0122_anyos_comp |
| 2 | MTFDKC83T8TFR | Drive_Bay | No reboot required | success | lnvgy_fw_drives_all.mu.7450-e2mu200-01e2_anyos_comp |
| 3 | ST1200MM0008 | Drive_Bay | No reboot required | success | lnvgy_fw_drives_all.stx.hdd.150-150d-00e2_anyos_comp |
=====

Package statistic results:
3 package(s) attempted
3 package(s) succeeded

Updatable Unit statistic results:
3 update(s) attempted
3 update(s) succeeded

Succeeded in running flash command
Succeed.
```

Configuration with OneCLI

On the BHS platform, OneCLI can be used to configure UEFI, BMC, and VPD. As OneCLI is switching to a Redfish-based interface, the command names for UEFI, BMC, and VPD will change.

Notes: With the exception of delete, all commands are supported on BHS.

Instance settings are expanded for BHS. To delete an instance of a setting, you need to enter the correct value for the specific setting. Example command:

```
OneCli.exe config set BMC.LoginId.2 "" -b userid:password@ipaddress
```

The following command shows the UEFI configuration – the screen capture is shown on the right:

```
./OneCli config show UEFI -b  
userid:password@ipaddress
```

```
Start to connect BMC at 10.240.197.164 to apply config show  
Invoking SHOW command ...  
UEFI.AdvancedRAS_DIMMDisablePolicy=DisableFaultyDIMMPersistently  
UEFI.AdvancedRAS_MachineCheckRecovery=Enabled  
UEFI.AdvancedRAS_PCIErrrorRecovery=Enabled  
UEFI.AdvancedRAS_PCIEEndpointResetOnFatalError=Disabled  
UEFI.BootModes_AcceleratedBoot=Disabled  
UEFI.BootModes_InfiniteBootRetry=Disabled  
UEFI.BootModes_PreventOSChangesToBootOrder=Disabled  
UEFI.BootModes_SystemBootMode=UEFI Mode  
UEFI.BootOrder_BootOrder_1=redhat  
UEFI.BootOrder_BootOrder_2=DVDROM  
UEFI.BootOrder_BootOrder_3=HardDisk  
UEFI.BootOrder_BootOrder_4=Network  
UEFI.BootOrder_BootOrder_5=USBStorage  
UEFI.BootOrder_DVDROMPriority_1=None  
UEFI.BootOrder_HardDiskPriority_1=HardDrive
```


Backing up and restoring the configuration with encryption

OneCLI supports an encrypted method to save and restore system configuration settings:

- `--passphrase`: The passphrase string to back up and restore configuration settings with a file
- `--scope`: Indicates the scope of the encrypted content – values: `WholeFile/SensitiveInfo`

Command:

```
OneCli.exe config backup --passphrase 'passphrasexxx' --bmc  
userid:password@ip --file backup.txt --scope wholefile
```

```
D:\work\build\BHS-500\lnvgy_utl_lxce_onecli01y-5.0.0_winsrv_x86-64>OneCli.exe config backup --passphrase cpassword --redfish  
-b USERID:LXCE_fvt@shcn2023@10.240.195.201 --file save.txt --scope wholefile  
[1s]Certificate check finished [100%][=====>]  
Start to connect BMC at 10.240.195.201 to apply config backup  
Invoking Backup command ...  
Backup succeed.  
Succeed.
```

Note: For ThinkSystem servers prior to the ThinkSystem V4 platform, the `-redfish` option is required to run this function.

New OneCLI FW update options

OneCLI has a new command option for bundle updates: `--cleanup`

This option works with the `update` command and is used to clear any existing previously staged task.

Example:

```
./OneCli update flash --dir pkg --bundle -b  
userid:password@ipaddress --applytime onstartupdaterequest --cleanup
```

```
[1m45s]Succeeded in downloading FPD logs [100%][=====]  
FPD logs has been saved to:3:\Tools\OneCli\lmgv_util\lxc_onecli82b-5.8.8_x86-64\logs\OneCli-20240722-155217-19348\
```

Flash Results:					
No.	Updatable Unit	Slot	Need Reboot	Result	Update ID
1	Firmware-PSoc:hx8B	PCI_Slot	No reboot required	Failure(*117)	lmgv_fw_sysboard_sep.g1-8B-11-4231_anyos_comp
2	Firmware-PSoc:hx8A	PCI_Slot	No reboot required	Failure(*117)	lmgv_fw_sysboard_sep.g1-8A-3-4231_anyos_comp
3	Broadcom 5719 1GbE RJ45 4-port OCP Ethernet Adapter	PCI_Slot 6	No reboot required	Failure(*117)	hrcn-lmgv_fw_nic_xl.14e4.1657.17aa.4384-227.8.3.1-4342_anyos_comp
4	LX8H	PCI_Slot	No reboot required	Failure(*117)	lmgv_fw_lxan_xal585c-1.18_anyos_comp

*Failure(*117) : Another update in progress.

Package statistic results:
4 update(s) attempted.
4 update(s) failed.

Screen capture of cleanup command is not used

```
Capture updates done, the result is stored to D:\Tools\OneCli\lmgv_util\lxc_onecli82b-5.8.8_x86-64\logs\OneCli-20240722-155217-19348\OneCli-update-compare.xml  
[1m]uploading finished [100%][=====]
```

Flash Results:					
No.	Updatable Unit	Slot	Need Reboot	Result	Update ID
1	Firmware-PSoc:hx8B	PCI_Slot	N/A	Staged(*46)	lmgv_fw_sysboard_sep.g1-8B-11-4231_anyos_comp
2	Firmware-PSoc:hx8A	PCI_Slot	N/A	Staged(*46)	lmgv_fw_sysboard_sep.g1-8A-3-4231_anyos_comp
3	Broadcom 5719 1GbE RJ45 4-port OCP Ethernet Adapter	PCI_Slot 6	N/A	Staged(*46)	hrcn-lmgv_fw_nic_xl.14e4.1657.17aa.4384-227.8.3.1-4342_anyos_comp
4	LX8H	PCI_Slot	N/A	Staged(*46)	lmgv_fw_lxan_xal585c-1.18_anyos_comp

*Staged(*46) : Pending and waiting user to execute OneCli update startstaged

Package statistic results:
4 update(s) attempted.
4 update(s) staged.

Updatable Unit statistic results:
4 update(s) attempted.
4 update(s) staged.

See the OneCli-update-Flash.xml and OneCli Logs in the output directory for details.

Task ID: 2
Use "OneCli update startstaged" to start the staged task
Use "OneCli update checktask --taskid <staged task id>" to check flashing result
* When the task start, target system may boot into iBFT setup and target system will reboot automatically after update task is completed

Some packages has been staged
Staged

Screen capture of using cleanup command

Miscellaneous new OneCLI functions – uefipassword

Command	Parameter	Description
uefipassword		Used to set or change power-on and administrator passwords
	--admin	Set or change the administrator password
	--poweron	Set or change the power-on password
	--oldpwd	Specify the old password
	--newpwd	Specify the new password

Example commands:

- **Creating a new UEFI admin password:**

```
- ./OneCli uefipassword --admin --newpwd "newpassword" -b  
USERID:PASSWORD@ipaddress
```

- **Updating a UEFI admin password:**

```
- ./OneCli uefipassword --admin --oldpwd "oldpassword" --newpwd  
"newpassword" -b USERID:PASSWORD@ipaddress
```

- **Deleting an old UEFI admin password:**

```
- ./OneCli uefipassword --admin --oldpwd "oldpassword" -b  
USERID:PASSWORD@ipaddress
```


Miscellaneous new OneCLI functions – multiserase

The new **multiserase** OneCLI command option is used to **securely erase disk drives** on **multiple servers**.

Example:

OneCli.exe multiserase --configfile sample/multi_task_config.json --erasefile sample/erasefile.json

// Supported **Secure Erase Case:**

connection_type	disk_type	erase_method	patterns
onboard	SATA	Normal, Enhanced	not supported
onboard	NVMe	Enhanced , Cryptographic	not supported
hba	SATA	Normal, Enhanced	not supported
hba	SAS	Cryptographic, Block	not supported
broadcom_raid	ALL	Simple , Normal , Thorough	00000000 ~ 11111111
broadcom_raid_sed	SED	Secure	not supported
microchip_raid	ALL	Zero , Random Zero , Random Random Zero, Crypto Scramble Sanitize , Block Erase Sanitize , Overwrite Sanitize	not supported
marvell_raid	NVMe	Enhanced , Cryptographic	not supported

Miscellaneous new OneCLI functions – edgserver

Command	Parameter	Description
edgserver		Used for edge server management
	getsecurityinfo	Get edge server security information from the BMC
	lock	Lock the server
	unlock	Unlock the server

Example command: `./OneCli edgserver getsecurityinfo -b USERID:PASSWORD@BMCIIP`

Click the button to see the sample output



getsecurityinfo

Miscellaneous new OneCLI functions – edgserver

```
./OneCli edgserver getsecurityinfo -b USERID:PASSWORD@BMCIP
```



```
MT:7DBY, SN:7DBYABC123
Active-code: 8BF3AD95AC337FA7
Active-status: Active
Mode: Disabled
Activation-method: Local
```

The Information from BMC:

Name	Value
Machine Type	7DBY
Serial Number	7DBYABC123
Activation Method	Local
Active Code	8BF3AD95AC337FA7
Public Key	86 49 0c 5d 64 36 0d af 2a 70 a4 7f 38 a9 3c d5 24 9d c8 b9 47 cd 2a 08 cf 6c 88 e0 5e 16 5f 0b e9 3a 57 50 2e 8f b1 a7 4b b1 38 51 26 ac 89 c3 5b 7e 62 5d af f e 9c 36 cb 99 a4 3d 88 4c 0d 66
Status	Active
Mode	Disabled
Challenge Text	
Device Id	DB AD 8D EA 71 C1 D3 2B 7A FE BD DC 70 1A 4B 48 14 87 24 EF 62 04 D3 28 90 5C EE 86 81 9F 41 FD
Motion Detection Enable	Disabled
Motion Detection ThresholdLevel	0_Steps
Motion Detection StepCounter	0
Chassis Intrusion Detection Enable	false
SED Encryption	false
SED Generate method	Unknown
User Role ID	Administrator+
Wave	Wave2

Succeed.



UpdateXpress ThinkEdge server security feature - 1

The other features can't be selected if **Configure security feature on ThinkEdge server** is selected.

Lenovo XClarity Essentials UpdateXpress v01s-5.0.0 Active Machine Type: TDBY

1. Welcome

2. Target Server

3. Task

4. Configure Security Fea...

5. Finish

Task
What task are you interested in?

- ☐ Perform updating on target server
Update the server selected at previous page. The updates will be acquired and automatically applied.
- ☐ Manage Staged Update
Start, cancel or list previously staged updates. This task only supports XCC2.
- ☐ Remote RAID configuration
Configure RAID controller using the BMC service.
- ☐ Backup System Configuration
Backup system configuration including Hardware and Firmware Inventory, VPD, FoD license, etc., to an external file.
- ☐ Restore System Configuration
Restore system configuration from backup you previously made.

☒ **Configure security feature on ThinkEdge server**
Configure security sensors, SED authentication keys, system lockdown control and activate locked server.

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UpdateXpress ThinkEdge server security feature - 2

While a ThinkEdge server is **unlocked**, these features are available for selection in the **ThinkEdge Server Security Feature** window.

Lenovo XClarity Essentials UpdateXpress v01s-5.0.0 Active Machine Type: TDBY

1. Welcome

2. Target Server

3. Task

4. Configure Security Features

5. Finish

ThinkEdge Server Security Feature

Select the tasks that you are interested in.

detection. When ThinkEdge server is in lockdown mode, it locks access to SED AK such that all access to data stored in SED drives is denied.

Depending on whether lockdown control is managed by XClarity Controller or ThinkShield Key Vault Portal, ThinkEdge server in lockdown mode can be activated by user with at least Administrator privilege in XCC, or user authenticated as owner of the server in ThinkShield Key Vault Portal.

You can learn more about ThinkEdge security in Lenovo Press at this [page](#).

I have read and understand these concepts

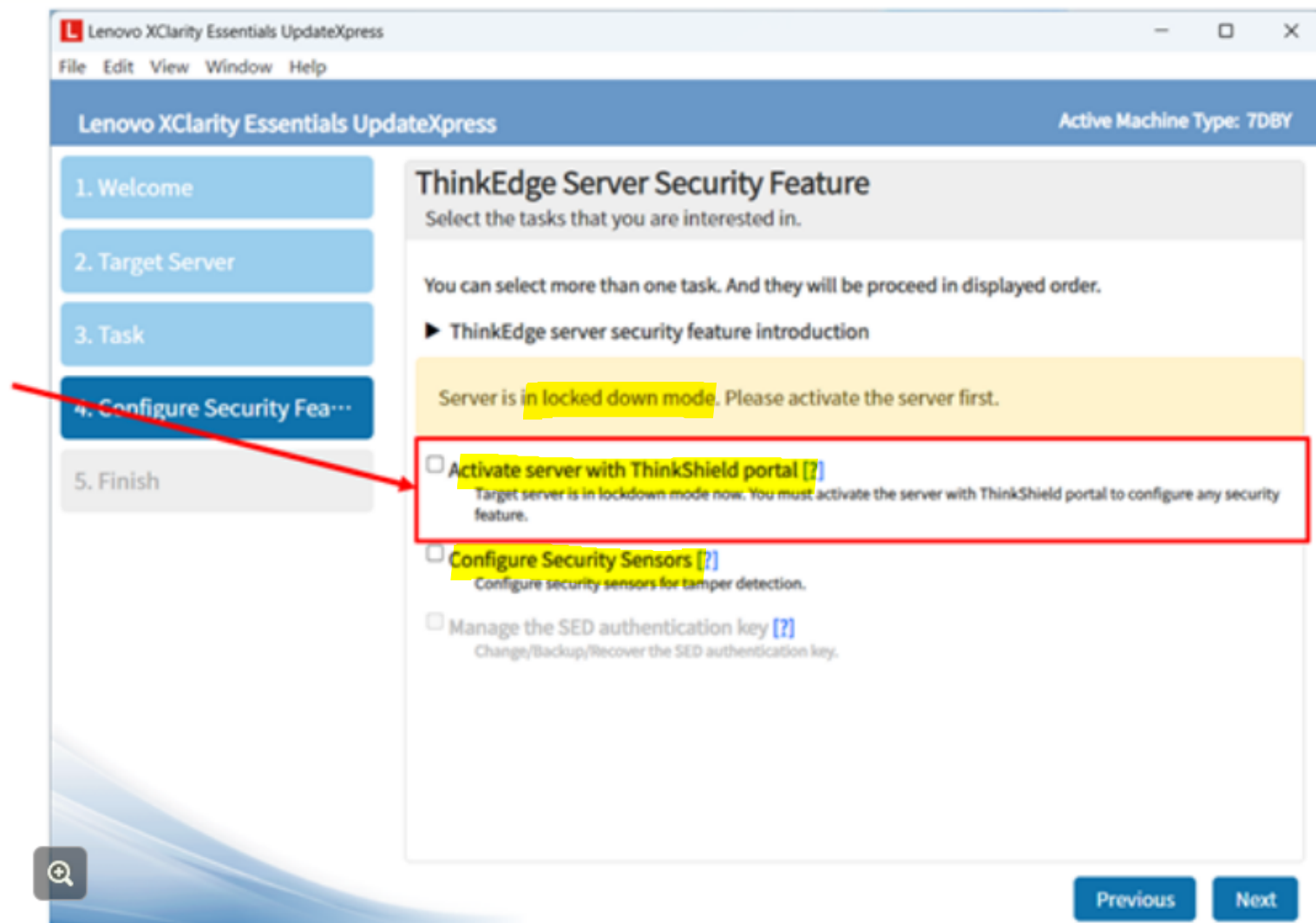
- ☐ **Claim server in ThinkShield portal** [?]
 Claim server's ownership in ThinkShield Key Vault Portal.
- ☐ **Configure Security Sensors** [?]
 Configure security sensors for tamper detection.
- ☐ **Manage the SED authentication key** [?]
 Change/Backup/Recover the SED authentication key.
- ☐ **System Lockdown Control** [?]
 Promote system lockdown control from XClarity Controller managed to ThinkShield portal managed.
- ☐ **Update Server's Public Key**
 Update public key for the current SystemBoard

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UpdateXpress ThinkEdge server security feature - 3

While a ThinkEdge server is locked, the **Activate server with ThinkShield Portal** feature will be available for selection in the **ThinkEdge Server Security Feature** window.

The server needs to be unlocked before another task can be performed.



Backing up and restoring the system configuration with UpdateXpress

These two new features are available for all ThinkSystem and ThinkEdge servers.

Lenovo XClarity Essentials UpdateXpress v01s-5.0.0 Active Machine Type: 7DBY

1. Welcome

2. Target Server

3. Task

4. Finish

Task

What task are you interested in?

- ☐ Perform updating on target server
Update the server selected at previous page. The updates will be acquired and automatically applied.
- ☐ Manage Staged Update
Start, cancel or list previously staged updates. This task only supports XCC2.
- ☐ Remote RAID configuration
Configure RAID controller using the BMC service.
- ☒ **Backup System Configuration**
Backup system configuration including Hardware and Firmware inventory, VPD, FoD license, etc., to an external file.
- ☒ **Restore System Configuration**
Restore system configuration from backup you previously made.
- ☐ Configure security feature on ThinkEdge server
Configure security sensors, SED authentication keys, system lockdown control and activate locked server.

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Backing up the system configuration with UpdateXpress

These items are optional.
The public key will also back up automatically if UpdateXpress is running on a ThinkEdge server.

Lenovo XClarity Essentials UpdateXpress v01s-5.0.0 Active Machine Type: TDBY

1. Welcome

2. Target Server

3. Task

4. Backup Config (1/2)

5. Finish

Backup System Configuration Setting

Backup file location:

D:\BHS\Invgy_utl_lxce_ux01s-5.0.0_windows_x86-64\backup.zip Browse...

Backup the following items:

- ☒ Hardware and Firmware Inventory
- ☒ Vital Product Data
- ☒ FoD License
- ☐ Firmware Configuration in XCC and UEFI
- ☒ SED Authentication Key

Set Password For SED Authentication Key

Note that you will be asked for this password when you use the file to restore a configuration.

Set Password Confirm Password

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New UpdateXpress features for ThinkEdge servers

The following **How to** videos demonstrate the new LXCE features for ThinkEdge servers. These videos can also be found on the **course landing page**.

- [How to manage SED AK using UpdateXpress](#)
- [How to claim a server in ThinkShield Portal using UpdateXpress](#)
- [How to upgrade lockdown control mode using UpdateXpress](#)
- [How to activate the server in lockdown mode using UpdateXpress](#)
- [How to configure security sensors using UpdateXpress](#)

BoMC support for the ST45 V3 FW update

Starting with BoMC 14.0.0, BoMC supports limited FW updates on the ST45 V3:

- No support for the BMC/XCC FW update
- The FW package is required

Support scope:

- UEFI
- ThinkSystem 4350-8i SAS/SATA 12Gb HBA
- ThinkSystem RAID 5350-8i PCIe 12Gb Adapter
- ThinkSystem Broadcom 5719 1GbE RJ45 4-Port PCIe Ethernet Adapter
- ThinkSystem Broadcom 57416 10GBASE-T 2-Port PCIe Ethernet Adapter
- Disk drive